

BEE 4750 Homework 5: Mixed Integer and Stochastic Programming

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Due Date

Thursday, 12/04/24, 9:00pm

Overview

Instructions

- In Problem 1, you will use mixed integer programming to solve a waste load allocation problem.
- In Problem 2, you will formulate a stochastic optimization problem.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
Activating project at `~/Desktop/Cornell_Courses/FA_25/BEE_4750/hw5-xindicliu`
Installed OpenBLAS32_jll      v0.3.29+0
Installed Measures            v0.3.3
Installed GR_jll              v0.73.18+0
```

```

Installed PlotUtils          v1.4.4
Installed OpenSSL            v1.6.0
Installed StaticArrays       v1.9.15
Installed MutableArithmetics v1.6.7
Installed JSON               v1.3.0
Installed HiGHS_jll          v1.12.0+0
Installed StaticArraysCore   v1.4.4
Installed GR                 v0.73.18
Installed Pango_jll          v1.57.0+0
Installed FFMPEG             v0.4.5
Installed METIS_jll          v5.1.3+0
Installed DataStructures     v0.19.3
Installed GraphRecipes       v0.5.15
Installed StatsBase          v0.34.8
Installed JuMP               v1.29.3
Installed StableRNGs         v1.0.4
Installed Adapt             v4.4.0
Installed HiGHS              v1.20.1
Installed FFMPEG_jll         v8.0.0+0
Installed ForwardDiff        v1.3.0
Installed StructUtils        v2.6.0
Precompiling project...
 443.0 ms Measures
 398.3 ms StaticArraysCore
 415.8 ms StableRNGs
 421.8 ms Adapt
 576.1 ms StructUtils
 502.2 ms METIS_jll
 908.5 ms ChainRulesCore
 447.5 ms OpenBLAS32_jll
1435.6 ms DataStructures
 694.5 ms ColorVectorSpace → SpecialFunctionsExt
1213.7 ms Pango_jll
1394.9 ms OpenSSL
1118.5 ms FFMPEG_jll
 529.1 ms DiffResults
 496.6 ms OffsetArrays → OffsetArraysAdaptExt
 682.0 ms Adapt → AdaptSparseArraysExt
3651.1 ms MutableArithmetics
3381.5 ms ColorSchemes
 646.5 ms StructUtils → StructUtilsTablesExt
 718.3 ms ChainRulesCore → ChainRulesCoreSparseArraysExt
 917.4 ms LogExpFunctions → LogExpFunctionsChainRulesCoreExt

```

1171.2 ms	SpecialFunctions → SpecialFunctionsChainRulesCoreExt
724.2 ms	SortingAlgorithms
2449.5 ms	JSON
4620.0 ms	StaticArrays
3113.7 ms	HiGHS_jll
3794.3 ms	FFMPEG
3035.9 ms	ForwardDiff
4545.8 ms	GR_jll
2048.0 ms	StatsBase
1284.7 ms	BenchmarkTools
619.8 ms	StaticArrays → StaticArraysChainRulesCoreExt
1267.6 ms	ArnoldiMethod
623.6 ms	StaticArrays → StaticArraysStatisticsExt
644.3 ms	Adapt → AdaptStaticArraysExt
1083.0 ms	ForwardDiff → ForwardDiffStaticArraysExt
7582.3 ms	PlotUtils
1645.5 ms	GeometryTypes
11725.9 ms	HTTP
2404.2 ms	Interpolations
3334.7 ms	Graphs
7804.6 ms	GeometryBasics
2271.5 ms	PlotThemes
1104.2 ms	Interpolations → InterpolationsForwardDiffExt
2811.0 ms	RecipesPipeline
2970.6 ms	GR
1281.9 ms	NetworkLayout
1699.2 ms	NetworkLayout → NetworkLayoutGraphsExt
2802.7 ms	GraphRecipes
29989.0 ms	DataFrames
26952.6 ms	MathOptInterface
1954.5 ms	Latexify → DataFramesExt
1782.9 ms	MathOptIIS
6850.4 ms	HiGHS
11905.4 ms	JuMP
37740.6 ms	Plots
3040.0 ms	Plots → GeometryBasicsExt

57 dependencies successfully precompiled in 66 seconds. 195 already precompiled.

```

using JuMP
using HiGHS
using DataFrames
using GraphRecipes

```

```
using Plots
using Measures
using MarkdownTables
```

Problems (Total: 30 Points)

Problem 1 (24 points)

Three cities are developing a coordinated municipal solid waste (MSW) disposal plan. Three disposal alternatives are being considered: a landfill (LF), a materials recycling facility (MRF), and a waste-to-energy facility (WTE). The capacities of these facilities and the fees for operation and disposal are provided below.

- **LF:** Capacity 200 Mg, fixed cost \$2000/day, tipping cost \$50/Mg;
- **MRF:** Capacity 350 Mg, fixed cost \$1500/day, tipping cost \$7/Mg, recycling cost \$40/Mg recycled;
- **WTE:** Capacity 210 Mg, fixed cost \$2500/day, tipping cost \$60/Mg;

The MRF recycling rate is 40%, and the ash fraction of non-recycled waste is 16% and of recycled waste is 14%. Transportation costs are \$1.5/Mg-km, and the relative distances between the cities and facilities are provided in the table below.

City/Facility	Landfill (km)	MRF (km)	WTE (km)
1	5	30	15
2	15	25	10
3	13	45	20
LF	-	32	18
MRF	32	-	15
WTE	18	15	-

The fixed costs associated with the disposal options are incurred only if the particular disposal option is implemented. The three cities produce 100, 90, and 120 Mg/day of solid waste, respectively, with the composition provided in the table below.

Component	% of total mass	Combustion ash (%)	MRF Recycling rate (%)
Food Wastes	15	8	0
Paper & Cardboard	40	7	55
Plastics	5	5	15
Textiles	3	10	10

Component	% of total mass	Combustion ash (%)	MRF Recycling rate (%)
Rubber, Leather	2	15	0
Wood	5	2	30
Yard Wastes	18	2	40
Glass	4	100	60
Ferrous	2	100	75
Aluminum	2	100	80
Other Metal	1	100	50
Miscellaneous	3	70	0

The information in the above table will help you determine the overall recycling and ash fractions. Note that the recycling residuals, which may be sent to either landfill or the WTE, have different ash content than the ash content of the original MSW. You will need to determine these fractions to construct your mass balance constraints.

Reminder: Use `round(x; digits=n)` to report values to the appropriate precision!

Problem 1.1

Based on the information above, calculate the overall recycling and ash fractions for the waste produced by each city.

Answer:

Quantity	Formula	Actual Calculation	Value
Overall recycling fraction	$r = \sum w_i r_i$	$0.15(0) + 0.40(0.55) + \dots + 0.03(0)$	0.3775 (37.75%)
Overall MSW ash fraction	$a_{\text{MSW}} = \sum w_i a_i$	$0.15(0.08) + 0.40(0.07) + \dots + 0.03(0.70)$	0.1641 (16.41%)

Problem 1.2

What are the decision variables for your optimization problem? Provide notation and variable meaning.

Answer:

Let:

$x_{i,j}$ = \$ mass flow (Mg/day) from city i to facility j , where $i \in \{1, 2, 3\}$ and $j \in \{LF, MRF, WTE\}$.

$y_{\{MRF,k\}}$ = \$ mass flow (Mg/day) of residue from MRF to facility k , where $k \in \{LF, WTE\}$.

$y_{\{WTE,LF\}}$ = \$ mass flow (Mg/day) of ash from WTE to LF.

δ_j = \$ binary variable indicating whether facility j is used, $j \in \{LF, MRF, WTE\}$, with $\delta_j \in \{0, 1\}$.

All mass-flow variables are nonnegative.

Problem 1.3

Formulate the objective function. Make sure to include any needed derivations or justifications for your equation(s).

Answer:

Parameter definitions: | Parameter | Meaning | | t | transport cost (\$/Mg · km) | | $d_{i,j}$ | distance from city i to facility j | | c_j^{tip} | tipping fee at facility j | | c^{rec} | recycling processing cost (\$/Mg recycled) | | r_{overall} | overall recycling fraction (=0.3775) | | r_{MRF} | MRF recycling rate (=0.40) | | F_j | fixed daily cost for facility j | | C_j | capacity of facility j | | M_j | big-M value for facility j (typically $M_j = C_j$) |

The objective is to minimize the total daily system cost:

$$Z = C_{\text{trans}} + C_{\text{tip}} + C_{\text{rec}} + C_{\text{fixed}}.$$

where - transport cost from cities to facilities:

$$C_{\text{trans},\text{city}} = t \sum_{i=1}^3 \sum_{j \in \{LF, MRF, WTE\}} d_{i,j} x_{i,j}.$$

- transport cost for inter-facility flows:

$$C_{\text{trans},\text{inter}} = t(d_{\text{MRF},\text{LF}} y_{\text{MRF},\text{LF}} + d_{\text{MRF},\text{WTE}} y_{\text{MRF},\text{WTE}} + d_{\text{WTE},\text{LF}} y_{\text{WTE},\text{LF}}).$$

1. Total transport cost

$$C_{\text{trans}} = C_{\text{trans},\text{city}} + C_{\text{trans},\text{inter}}.$$

2. Tipping fees

$$C_{tip} = c_{LF}^{tip} \left(\sum_{i=1}^3 x_{i,LF} + y_{MRF,LF} + y_{WTE,LF} \right) + c_{MRF}^{tip} \left(\sum_{i=1}^3 x_{i,MRF} \right) + c_{WTE}^{tip} \left(\sum_{i=1}^3 x_{i,WTE} + y_{MRF,WTE} \right).$$

3. MRF recycling costs

$$C_{rec} = c^{rec} r_{MRF} \left(\sum_{i=1}^3 x_{i,MRF} \right).$$

4. Fixed facility operating costs

$$C_{fixed} = F_{LF} \delta_{LF} + F_{MRF} \delta_{MRF} + F_{WTE} \delta_{WTE}.$$

Putting all components together:

$$Z = C_{trans} + C_{tip} + C_{rec} + C_{fixed}.$$

Problem 1.4

Derive all relevant constraints. Make sure to include any needed justifications or derivations.

Answer:

City mass balance (for $i = 1, 2, 3$):

$$\sum_{j \in \{LF, MRF, WTE\}} x_{i,j} = S_i.$$

MRF inflow and split:

$$X_{MRF} = \sum_{i=1}^3 x_{i,MRF}.$$

$$M^{rec} = r_{MRF} X_{MRF}, \quad M^{res} = (1 - r_{MRF}) X_{MRF}.$$

MRF residual routing:

$$y_{MRF,LF} + y_{MRF,WTE} = (1 - r_{MRF}) \sum_{i=1}^3 x_{i,MRF}.$$

WTE inflow:

$$X_{WTE} = \sum_{i=1}^3 x_{i,WTE} + y_{MRF,WTE}.$$

WTE ash:

$$y_{WTE,LF} = 0.1641 \sum_{i=1}^3 x_{i,WTE} + 0.1386 y_{MRF,WTE}.$$

Landfill inflow:

$$X_{LF} = \sum_{i=1}^3 x_{i,LF} + y_{MRF,LF} + y_{WTE,LF}.$$

Capacity constraints:

$$\sum_{i=1}^3 x_{i,MRF} \leq C_{MRF},$$

$$\sum_{i=1}^3 x_{i,WTE} + y_{MRF,WTE} \leq C_{WTE},$$

$$\sum_{i=1}^3 x_{i,LF} + y_{MRF,LF} + y_{WTE,LF} \leq C_{LF}.$$

Big-M linking constraints:

(to enforce zero flow when facility closed)

$$\sum_{i=1}^3 x_{i,MRF} \leq M_{MRF} \delta_{MRF},$$

$$\sum_{i=1}^3 x_{i,WTE} + y_{MRF,WTE} \leq M_{WTE} \delta_{WTE},$$

$$\sum_{i=1}^3 x_{i,LF} + y_{MRF,LF} + y_{WTE,LF} \leq M_{LF} \delta_{LF}.$$

- Choose $M_j \geq C_j$, typically $M_j = C_j$.

Nonnegativity and integrality:

$$x_{i,j} \geq 0, y_{MRF,LF} \geq 0, y_{MRF,WTE} \geq 0, y_{WTE,LF} \geq 0.$$

$$\delta_{LF}, \delta_{MRF}, \delta_{WTE} \in \{0, 1\}.$$

Problem 1.5

Find the optimal solution (using JuMP to solve the problem). Report the optimal objective value.

```
# Data
S = [100.0, 90.0, 120.0]
C_LF = 200.0; C_MRF = 350.0; C_WTE = 210.0

# MRF recycling rate = 40%
r_MRF = 0.40

# Ash fractions from problem 1.1
a_MSW = 0.1641
a_res = 0.1386

# Transportation cost ($/Mg-km)
t = 1.5

# Distances (km)
d_city_LF = [5.0, 15.0, 13.0]
d_city_MRF = [30.0, 25.0, 45.0]
d_city_WTE = [15.0, 10.0, 20.0]
d_MRF_LF = 32.0; d_MRF_WTE = 15.0; d_WTE_LF = 18.0

# Tipping fees ($/Mg)
c_tip_LF = 50.0; c_tip_MRF = 7.0; c_tip_WTE = 60.0

# Recycling cost ($/Mg recycled)
c_rec = 40.0

# Fixed costs ($/day)
F_LF = 2000.0; F_MRF = 1500.0; F_WTE = 2500.0

# Big-M values (choose tight = capacities)
M_LF = C_LF; M_MRF = C_MRF; M_WTE = C_WTE

# Model
model = Model(HiGHS.Optimizer)
set_silent(model)

# Variables
@variable(model, x[i = 1:3, j = 1:3] >= 0)
```

```

@variable(model, y_MRF_LF >= 0)
@variable(model, y_MRF_WTE >= 0)
@variable(model, y_WTE_LF >= 0)
@variable(model, delta_LF, Bin)
@variable(model, delta_MRF, Bin)
@variable(model, delta_WTE, Bin)

# City mass balance constraints
@constraint(model, [i = 1:3], sum(x[i, j] for j = 1:3) == S[i])

# MRF residue split for non-recycled portion
@constraint(model, y_MRF_LF + y_MRF_WTE == (1 - r_MRF) * sum(x[i, 2] for i = 1:3))

# WTE ash
@constraint(model, y_WTE_LF == a_MSW * sum(x[i, 3] for i = 1:3) + a_res * y_MRF_WTE)

# Capacity constraints
@constraint(model, sum(x[i, 1] for i = 1:3) + y_MRF_LF + y_WTE_LF <= C_LF)
@constraint(model, sum(x[i, 2] for i = 1:3) <= C_MRF)
@constraint(model, sum(x[i, 3] for i = 1:3) + y_MRF_WTE <= C_WTE)

# Big-M linking constraints
@constraint(model, sum(x[i, 1] for i = 1:3) + y_MRF_LF + y_WTE_LF <= M_LF * delta_LF)
@constraint(model, sum(x[i, 2] for i = 1:3) <= M_MRF * delta_MRF)
@constraint(model, sum(x[i, 3] for i = 1:3) + y_MRF_WTE <= M_WTE * delta_WTE)

# Objective: Minimize total cost
transport_city = t * (
    sum(d_city_LF[i] * x[i, 1] for i = 1:3) +
    sum(d_city_MRF[i] * x[i, 2] for i = 1:3) +
    sum(d_city_WTE[i] * x[i, 3] for i = 1:3)
)

transport_inter = t * (d_MRF_LF * y_MRF_LF + d_MRF_WTE * y_MRF_WTE + d_WTE_LF * y_WTE_LF)

tip_cost = c_tip_LF * (sum(x[i, 1] for i = 1:3) + y_MRF_LF + y_WTE_LF) +
    c_tip_MRF * (sum(x[i, 2] for i = 1:3)) +
    c_tip_WTE * (sum(x[i, 3] for i = 1:3) + y_MRF_WTE)

recycle_cost = c_rec * r_MRF * sum(x[i, 2] for i = 1:3)

fixed_cost = F_LF * delta_LF + F_MRF * delta_MRF + F_WTE * delta_WTE

```

```

@objective(model, Min, transport_city + transport_inter + tip_cost + recycle_cost + fixed_co

# Solve
optimize!(model)

# Output results
status = termination_status(model)
println("Status: ", status)

if status == OPTIMAL
    println("Objective value: \$", round(objective_value(model); digits = 2))

    println("\nFlows (x[i, j]) where j = 1:LF, 2:MRF, 3:WTE:")
    for i in 1:3, j in 1:3
        v = value(x[i, j])
        if v > 1e-6
            println("x[$i, $j] = ", round(v; digits = 3))
        end
    end

    println("\ny_MRF_LF = ", round(value(y_MRF_LF); digits = 3))
    println("y_MRF_WTE = ", round(value(y_MRF_WTE); digits = 3))
    println("y_WTE_LF = ", round(value(y_WTE_LF); digits = 3))

    println("\ndelta_LF = ", Int(value(delta_LF)))
    println("delta_MRF = ", Int(value(delta_MRF)))
    println("delta_WTE = ", Int(value(delta_WTE)))
end

```

```

Status: OPTIMAL
Objective value: $27855.48

```

```

Flows (x[i, j]) where j = 1:LF, 2:MRF, 3:WTE:

```

```

x[1, 1] = 100.0
x[2, 3] = 90.0
x[3, 1] = 78.405
x[3, 3] = 41.595

```

```

y_MRF_LF = 0.0
y_MRF_WTE = 0.0
y_WTE_LF = 21.595

```

$\text{delta_LF} = 1$
 $\text{delta_MRF} = 0$
 $\text{delta_WTE} = 1$

Answer:

The optimal solution is WTE takes all the 90 Mg of wastes from City 2, then another 41.595 Mg from City 3. It will generate about 21.595 Mg ashes that will then be transported to LF. More than the ashes, LF also takes 100 Mg of wastes from City 1 and 78.405 Mg of wastes from City 3. This gives MTE 131.595 Mg of total wastes to treat, and 200 Mg afterwards in total for LF, while MRF is completely unused. The optimal objective value is \$27855.48.

Problem 1.6

Draw a diagram showing the flows of waste between the cities and the facilities. Which facilities (if any) will not be used? Does this solution make sense?

Answer:

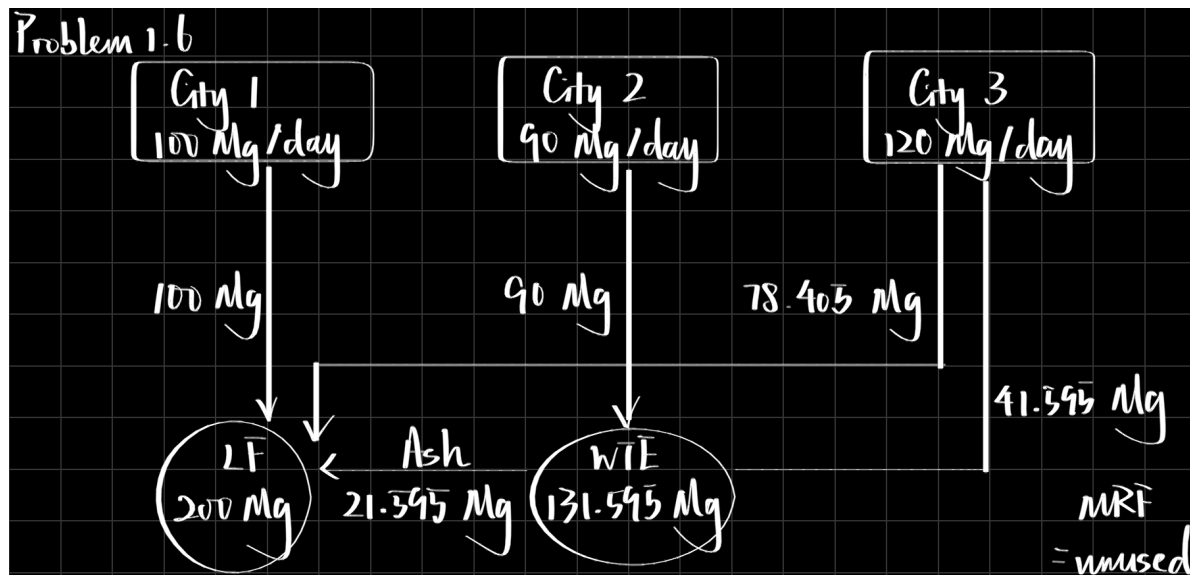


Figure 1: Description

The facility that will not be used is the MRF. This solution makes sense because using the MRF incurs not only a fixed daily cost of \$1500, but also a processing cost of \$40/Mg recycled. With a low tipping fee of \$7/Mg, the savings from sending waste through MRF versus other alternatives is small, so the combined costs outweigh the savings here. In addition, with the LF

capacity available up to 200 Mg and WTE available up to 210 Mg, the model can meet demand without paying the MRF costs. Thus, not using MRF is economically and physically consistent given the data, with capacity limits satisfied, mass conserved, and objective minimized.

Problem 2 (6 points)

Consider a two-period economic dispatch problem, based on the multi-period example from Lecture 14 (on 10/29). The generator data, including ramping constraints for each generator, is provided in ‘data/generators.csv.’ In period 1, the demand is $d_1 = 1100\text{MW}$. In period 2, the demand is projected to be $d_2 = 1200\text{MW}$, but there is a 25% probability that it is \$1500 . In the first period, the solar capacity factor is 0.9 and the wind capacity factor is 0.45, but in the second period, there is some uncertainty: the forecasted solar and wind capacity factors are 0.95 and 0.4, respectively, but there is a 30% probability that they are 0.75 and 0.5. Your goal is to identify how to dispatch your generators to minimize the cost of meeting demand.

Problem 2.1

Draw a scenario tree for this problem.

Answer:

Problem 2.1

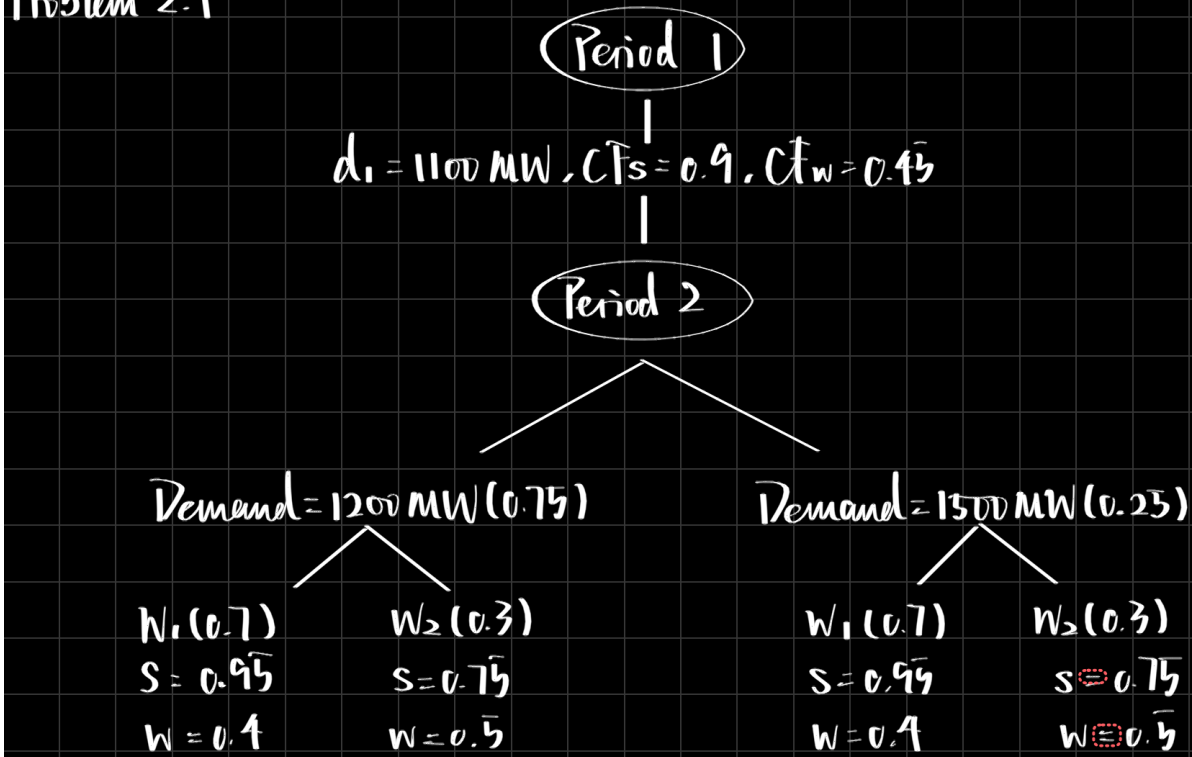


Figure 2: Description

Problem 2.2

Formulate a stochastic linear program for this problem based on your scenario tree from Problem 2.1 and the data in `data/generators.csv`.

Answer:

Sets: - Generators $i \in \mathcal{G}$. - Time periods $t = 1, 2$. - Scenarios $s \in \mathcal{S} = \{1, 2, 3, 4\}$.

Parameters: - $P_i^{\min}, P_i^{\max}$ = min/max output of generator i . - c_i = variable cost of generator i . - R_i = ramp limit of generator i . - $d_1 = 1100$. - $d_2^s \in \{1200, 1500\}$ depending on scenario. - Solar/wind capacity factors: - Period 1: $CF_{\text{solar}}^{(1)} = 0.9$, $CF_{\text{wind}}^{(1)} = 0.45$. - Period 2: CF_{solar}^s , CF_{wind}^s . - Scenario probabilities: - $p_1 = 0.525$, $p_2 = 0.225$, $p_3 = 0.175$, $p_4 = 0.075$.

Decision variables: - $g_{i,1}$ = generation of unit i in period 1. - $g_{i,2}^s$ = generation of unit i in period 2 under scenario s .

Objective:

$$\min \sum_i c_i g_{i,1} + \sum_s p_s \sum_i c_i g_{i,2}^s$$

Constraints:

1. Period-1 capacity limits:

$$P_i^{\min} \leq g_{i,1} \leq P_i^{\max}.$$

2. Period-1 renewable availability:

$$g_{\text{wind},1} \leq 0.45 P_{\text{wind}}^{\max}, \quad g_{\text{solar},1} \leq 0.90 P_{\text{solar}}^{\max}.$$

3. Period-1 demand:

$$\sum_i g_{i,1} = 1100.$$

4. Period-2 capacity limits:

$$P_i^{\min} \leq g_{i,2}^s \leq P_i^{\max} \quad \forall s.$$

5. Period-2 renewable availability:

$$g_{\text{wind},2}^s \leq \text{CF}_{\text{wind}}^s P_{\text{wind}}^{\max}, \quad g_{\text{solar},2}^s \leq \text{CF}_{\text{solar}}^s P_{\text{solar}}^{\max} \quad \forall s.$$

6. Period-2 demand (scenario-dependent):

$$\sum_i g_{i,2}^s = d_2^s \quad \forall s.$$

7. Ramping:

$$-R_i \leq g_{i,2}^s - g_{i,1} \leq R_i \quad \forall i, s.$$

8. Nonnegativity:

$$g_{i,1} \geq 0, \quad g_{i,2}^s \geq 0.$$

References

List any external references consulted, including classmates.

Used ChatGPT for problems 1.3, 1.4, 2.2, for the purpose of helping with LaTeX syntax, since it is really complicated to utilize with all the different symbols and versions. OpenAI. (2025). ChatGPT (Feb 1 GPT-4 model) [Large language model]. <https://chat.openai.com>